

INVERELL, Australia

WMO: 945410

Lat: 29.7753S

Lon: 151.0819E

Elev: 664

StdP: 93.60

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.0	2.3	-8.2	2.0	16.9	-6.1	2.4	15.2	6.5	16.5	6.1	15.0	1.4	180	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	33.9	18.8	32.3	18.6	30.8	18.3	22.0	28.2	21.1	27.6	20.4	26.6	3.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	16.4	23.9	19.4	15.3	23.1	18.5	14.5	22.7	68.0	28.3	64.4	27.6	61.8	26.6	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.2	5.4	-2.3	37.0	1.2	2.7	-3.1	38.9	-3.8	40.4	-4.4	41.9	-5.3	43.8		
			-1.2	26.0	0.8	3.7	-1.8	28.7	-2.3	30.9	-2.7	33.0	-3.3	35.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	23.2	22.9	20.9	17.9	13.3	10.8	9.9	10.9	14.1	17.4	20.1	21.8
	DBStd	5.45	2.50	2.54	2.25	2.25	2.57	2.57	2.38	2.65	2.90	3.12	3.11	2.85
	HDD10.0	83	0	0	0	0	6	22	32	20	2	1	0	0
	HDD18.3	1118	1	1	4	32	156	227	263	231	131	53	15	5
	CDD10.0	2605	411	362	338	238	108	46	27	48	126	231	304	366
	CDD18.3	597	153	129	84	20	0	0	0	1	4	25	69	113
	CDH23.3	5520	1410	1178	637	160	3	0	0	6	52	281	721	1072
	CDH26.7	1888	558	456	157	14	0	0	0	2	7	49	249	397
Wind	WSAvg	2.7	3.0	3.0	2.8	2.6	2.4	2.5	2.4	2.6	2.7	2.8	3.0	3.0
Precipitation	PrecAvg	767	102	81	62	48	54	33	50	45	43	70	78	98
	PrecMax	1049	266	267	175	219	180	90	174	154	238	169	181	313
	PrecMin	434	4	19	2	0	0	1	0	0	0	17	6	10
	PrecStd	167	69	61	49	49	45	24	37	35	41	40	43	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	35.9	32.1	28.4	23.7	19.6	19.9	23.0	27.6	30.7	34.9	35.0
		MCWB	19.6	19.2	18.4	16.2	13.5	12.0	11.6	12.0	16.5	15.2	18.1	18.6
	2%	DB	33.5	33.5	30.1	26.5	22.0	18.4	18.1	20.5	24.9	28.1	32.0	33.0
		MCWB	19.4	19.3	18.2	15.7	13.1	11.5	10.7	10.8	13.8	15.0	17.3	18.2
	5%	DB	31.7	31.5	28.4	25.2	20.6	17.3	16.6	18.8	22.7	26.5	29.6	31.0
		MCWB	19.2	19.0	17.8	15.4	12.2	11.0	10.0	10.1	12.7	15.0	16.6	17.7
	10%	DB	29.9	29.5	26.8	23.8	19.3	16.0	15.3	17.2	20.8	24.7	27.6	28.9
		MCWB	18.9	18.7	17.6	14.9	11.7	10.3	9.6	9.5	12.2	14.8	16.3	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.5	23.1	21.8	18.6	16.1	14.5	15.2	14.9	17.7	20.1	20.7	22.0
		MCDB	29.2	30.0	26.5	24.2	20.4	16.4	16.6	20.9	24.4	24.5	29.7	28.3
	2%	WB	22.1	21.8	20.4	17.5	14.8	13.4	13.2	12.9	15.7	17.8	19.4	20.7
		MCDB	27.6	28.9	25.7	22.7	18.4	15.6	15.3	16.3	21.1	23.4	27.0	27.7
	5%	WB	21.2	21.0	19.6	16.6	13.8	12.5	11.6	11.8	14.6	16.8	18.6	19.9
		MCDB	27.5	27.3	24.5	22.2	17.9	15.1	14.5	16.4	19.4	23.0	25.3	26.4
	10%	WB	20.5	20.1	18.8	15.8	12.8	11.5	10.4	10.7	13.5	15.9	17.9	19.2
		MCDB	26.7	25.9	23.7	21.7	17.6	14.8	13.9	15.7	18.3	22.3	24.5	25.1
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.8	11.3	11.5	11.6	9.8	10.7	12.1	12.5	12.4	12.5	12.1
		MCDBR	14.7	14.8	13.4	13.2	13.0	11.2	12.4	14.1	14.9	14.7	15.3	15.1
		MCWBR	4.6	4.5	4.5	4.7	5.5	5.5	6.3	6.8	6.9	6.0	5.3	5.0
	5% WB	MCDWR	11.1	11.4	10.5	11.1	10.0	8.3	8.9	10.7	11.5	12.2	12.0	11.4
	MCWBR	4.8	4.5	4.9	4.8	5.2	5.0	5.6	6.2	6.3	6.0	4.9	5.0	
Clear-Sky Solar Irradiance	taub	0.339	0.337	0.314	0.297	0.276	0.272	0.267	0.278	0.312	0.331	0.347	0.344	
	taud	2.580	2.600	2.663	2.665	2.676	2.690	2.688	2.631	2.516	2.495	2.501	2.540	
	Ebn at Noon	1001	986	979	948	925	908	928	955	964	980	987	999	
	Edh at Noon	106	101	89	80	71	66	69	81	101	111	114	111	
All-Sky Solar Radiation	RadAvg	7.18	6.48	5.55	4.64	3.69	3.01	3.38	4.33	5.52	6.36	6.80	7.14	
	RadStd	0.48	0.53	0.38	0.24	0.22	0.26	0.25	0.31	0.44	0.53	0.66	0.59	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+228	+139

Nomenclature: See separate page